

The main estimator families are direct-measurement methods, model-based observers, data-driven models, and combinations of these approaches. Figure 3.1 summarizes this relationship. Coulomb counting is the principal direct method. Equivalent-circuit or electrochemical models provide the physical basis for observers. Neural networks learn a nonlinear mapping from measured and engineered variables to the target state. Hybrid estimators combine physical propagation with learned capacity, parameter, or state correction.[24, 25, 26, 27]

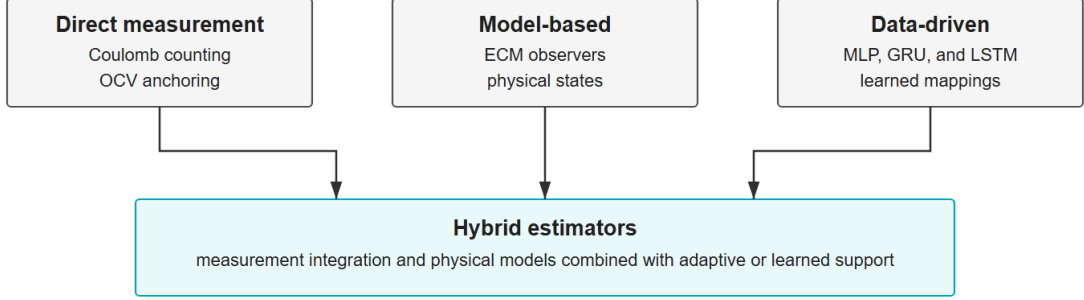


Figure 3.1: Main families of battery-state estimators. Hybrid methods combine elements of direct measurement, physical model structure, and data-driven adaptation.

3.3 Model-Based Estimation and Kalman Filtering

Equivalent-circuit models (ECMs) represent the terminal behaviour of a cell through an open-circuit-voltage source, an ohmic resistance, and one or more resistor-capacitor branches. The SOC and polarization voltages form the model state. Measured current is the input, while the predicted terminal voltage is compared with the measured voltage. This structure is computationally efficient and is therefore widely used in practical BMS observers.[4, 28, 29]

A nonlinear discrete state-space model can be written as

$$\mathbf{x}_k = f(\mathbf{x}_{k-1}, \mathbf{u}_k) + \mathbf{w}_k, \quad (3.2)$$

$$\mathbf{z}_k = h(\mathbf{x}_k, \mathbf{u}_k) + \mathbf{v}_k, \quad (3.3)$$

where $\mathbf{x}_k$ contains the hidden battery states, $\mathbf{u}_k$ contains known or measured inputs, and $\mathbf{z}_k$ contains the sensor measurements used for correction. In a battery observer, $\mathbf{u}_k$ commonly contains the cell current and $\mathbf{z}_k$ the terminal voltage. The